

A Review on Polymer Nanocomposites and Its Applications

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ABSTRACT: Multiple phases that have at least one, two, or three dimensions in the nanoscale range make up a nanocomposite material. Phase interfaces are produced when material dimensions are reduced to the nanoscale level, and they are crucial for improving the properties of the material. Understanding the relationship between structure and property is directly influenced by the surface area to volume ratio of the reinforced material employed during the creation of nanocomposites. Opportunities for overcoming barriers in the medical, pharmaceutical, food packaging, electronics, and energy industries are provided by nanocomposites on entirely new scales. Polymeric nanocomposites are frequently created by directly incorporating inorganic nanoscale building pieces into polymers. Surface modification of the nanofillers is crucial in the creation of nanocomposites through dispersive blending. It has the power to make fillers more hydrophobic, improve interfacial adhesion by chemical bonding or chain entanglement, and get rid of the loose structure of filler agglomerates. The state of the art for nanoparticle/polymer composites is reviewed in the current work, along with the specific surface pretreatment methods and their uses. Particular emphasis is placed on the function of modified nanoparticles and the mechanisms involved in the enhancement of the mechanical characteristics and wear resistance of the composites.

KEYWORDS Nanocomposites, polymer matrix composites, surface modification, mechanical properties, Tribological performance

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INTRODUCTION

Nanotechnology is a rapidly expanding field of study that deals with very small-scale particles and has a wide range of academic applications, as well as innovation, development, and commercial activities. In general, it has swamped fields including physics, chemistry, biology, material science, and engineering. This technology is therefore referred to as transdisciplinary technology. Its dimensions or size can specify its use and ideas [1, 2]. Due to its numerous applications in the fields of energy production and storage, information technology, medical applications, equipment's, food, bases of equipment, and water purification, as well as applications in the biomedical field and environmental issues, nanotechnology has the potential to transform daily life. The market offers many different product varieties including components for electrical devices, paints, energy storage, textiles, and medical equipment. The name alone indicates how tiny the particle is. It offers

a high surface area to volume ratio because of the size of the materials. On the material's surface, there are few imperfections visible. As a result, the qualities of nanomaterial are enhanced.

Materials with at least one dimension of the nanometer are known as nanocomposites. It demonstrates top-notch performance and the peculiar properties and distinctive design options of composite material combinations. They are made up of two or more different composites, each with unique physical and chemical properties, and most often, different phases are separated by an interface. As a result, no one of the individual ingredients can produce a nanocomposite with unique features. The matrix refers to the larger number of components. The process of adding components to the matrix material to improve the properties of nanocomposites is known as reinforcing. The equivalent of a nano-filler substance is a

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reinforcement or nanomaterial. Due to their ingredients, nanocomposites exhibit anisotropic characteristics and inhomogeneous reinforcement. Alternatives to micro and monolithic composites' stoichiometry and elemental compositional control restrictions are nanocomposite materials. According on how they are made, nanocomposites are categorized. Different kinds of nanoparticles and matrix materials are employed in the development of nanocomposite. According on the host or matrix material utilized, there are three groups of nanocomposites [3]. There are three types of them: Polymer Matrix Nanocomposites (PMNC), Metal Matrix Nanocomposites (MMNC), and Ceramic Matrix Nanocomposites (CMNC).

Ceramic matrix nanocomposites are cutting-edge engineering materials with a variety of uses in the mechanical and electrical fields. Different synthetic techniques for the creation of ceramic matrix nanocomposites (CMNC) have been published in the literature. The described preparation methods include the spray pyrolysis approach, the polymer precursor route, and the powder method. Chemical approaches include the sol-gel technique, the colloidal route and precipitation method, and template synthesis [4]. There are reports of many CMNC materials, including CNT-based materials[5].

By encrusting nanoparticles in an alloy matrix, Metal Matrix Nanocomposites (MMNC) are created. These composites are made of nanoparticles and a metal or metal alloy matrix. The intriguing aspect is that this nanomaterial has a variety of applications in electronics, equipment, LED, storage devices, and other areas because of its unique features compared to the host matrix. The characteristics of various nanoparticles vary. The properties of a particle will vary if its size does. By adding these particles to the matrix material, mechanical properties, damping characteristics, and resistance properties can all be improved. The common preparation processes include pyrolysis, liquid metal infiltration, vapor deposition, solidification, and chemical approaches such colloidal and sol-gel procedures [4]. There have been reports of several metal matrix nanocomposites [6].

Nanomaterials comprised of polymer nanocomposites and nanoparticles are called nanomaterials. Monomers are combined to form polymers. Polymer nanocomposites are

made up of inorganic nanoparticles that are distributed throughout a polymer matrix [7]. Through the synergistic interaction of the two constituent materials, these structures integrate the greatest qualities of each component [8]. These include enhanced electrical, ionic, mechanical, chemical, and other properties that can be used in a variety of ways. The materials in the nano area with the widest range of uses are polymer nanocomposites. It can be created by mixing inorganic particles with various qualities with an organic polymer matrix's processability and flexibility. Polymer nanocomposites are made of a polymer matrix that has been implanted with nanoscale components. The combination of monomers with flexible, lightweight, and affordable production characteristics results in polymers. Magnetic, mechanical catalytic and electrical properties can be found in inorganic nanoparticles. Both can be combined to create new material with a variety of applications.

PREPARATION OF NANOCOMPOSITES

There are various techniques used to prepare polymer nanocomposites. These include the Solution Cast Method, Spin Coating Method, Hot Press and Dip Coating Route, Melt Intercalation Method, and In Situ Polymerization Method.

Solution Cast Technique: This is the simplest traditional method for producing polymer film in a range of thicknesses. First, separate preparations of the nanoparticle solution and polymer salt are required. The necessary quantity of the polymer host is dissolved in the designated solvent by continuous stirring to produce polymer salt. The homogeneous polymer matrix is then given the required amount of salt, which is then mixed in until a homogeneous, transparent solution is obtained. Solution nanoparticles are introduced to a solvent during the manufacture of nanoparticles in order to achieve better dispersion. The produced nanoparticle solution is now added to the polymer salt solution, and the mixture is agitated until a homogeneous solution is obtained, resulting in the polymer nanocomposite. In order to allow the solvent to evaporate, the viscous solution was lastly kept in the petri dish at ambient temperature. The remaining solvent residues are then cleaned out of the solution and the petri dish coating is peeled off [9].

Spin Coating: The spin coating process is a quick and easy way to make homogenous films that range in thickness from a few nanometers to a few microns. This approach uses a spin coater to disperse a little amount of solvent at a predetermined speed on the substrate. When heated to evaporate the solvent, the mixture on a substrate spreads through centripetal acceleration. The major goal is to maintain the substrate's perpendicularity to the spinning axis while coating thin films on it. The parameters (i) mixture viscosity (ii) concentration (iii) rotational speed (iv) and spin time (v) can alter the film thickness. Only low viscosity solution mixtures are appropriate for this technique. For higher order viscosity mixtures, it is inapplicable. In order to create a thin film, the mixture cannot be spread by the spin coater rotation if it is of the gel type [10, 11].

Hot Press Technique: This is a quick and inexpensive way. This method, which uses no solvent, creates superior thin films. Dense materials are used to make these films. This method includes a heating chamber with a temperature controller, a basement, and a cylinder for weighing the sample. Most often, polymers are employed using this approach. The agate pestle is filled with the polymer, nanoparticle, and solvent. After the requisite amount of time, the combination is heated to a temperature that is roughly close to the melting point of the polymer and the slurry that was produced. Between the stainless steel blocks is the sample that was acquired. The system's pressure on the slurry is what creates the film [12, 13].

Dip Coating: The Dip Coating method produces high-quality film on both sides of the substrate. The method's minimal cost makes it special. There are three steps in it: immersion, deposition, and drainage. Afterward, the solvent evaporates. The substrate is dipped into the solution in the first stage at a rate that will give enough time for coating. The following process entails deposition and drainage. The substrate is kept submerged in the solvent after that. It will be necessary for the solvent to interact with the substrate. Now, the substrate is removed slowly and steadily to aid in the formation of thin coating on the substrate. Evaporation, which includes the solvent, is the final phase. The substrate is then heated to eliminate the excess solvent that has already built up after this has evaporated. The solution's density and

removal rate both affect the film's quality and thickness [14].

Melt Intercalation Technique: Because it is inexpensive, environmentally benign, and requires no solvent, it is a crucial technology. The key benefit of this method is that it will improve the thermal characteristics and result in greater nanoparticle dispersion. The nanoparticle's surface characteristics could change and it could deteriorate due to the high temperature. To achieve uniform dispersion, the polymer matrix is first heated to a high temperature and then added the nanoparticles. This process is then thoroughly mixed for a predetermined amount of time. Dispersion has the power to regulate processing conditions, host polymer compatibility, and surface modification of nanofiller [15, 16].

In-Situ Polymerization Method: This broad technique is used to create polymer nanocomposites. In this procedure, a suitable solvent is used to combine the monomer with the nanoparticles. Simply put, to create a polymer nanocomposite, monomers are intercalated with nanoparticles and then polymerized with an appropriate reagent or free radical initiator. It is founded on a bottom-up strategy. With the aid of this technique, multidimensional, clearly defined structures can be created that have entirely new attributes from their original ancestors. In a polymeric matrix, uniform dispersion is typically challenging. This method is used to accomplish that. In the polymer matrix, the metal precursor is employed to manufacture metal or metal oxide particles from nanoparticles. Particle size and morphology can be controlled using the in-situ approach [17]. The most popular method for converting metal nanoparticles into organic phase chemically is the sol-gel process [18].

Surface Pretreatment Techniques

Surface modification of inorganic fillers often entails the application of organic coatings, which can be accomplished by utilizing physical and chemical interaction between the fillers and the modifiers. The resulting bonding is frequently a combination of chemical and secondary connections. For microcomposites, a great deal of knowledge has been gained regarding processing methods, interfacial characterization, and the impact of interfacial properties on composites [19]. It is well known that there is no discernible difference between

the treatments of microparticles and nanoparticles, other than the fact that the modifiers find it very challenging to enter the tightly bound nanoparticle agglomerates, particularly in the case of physical treatments like surfactant treatment and polymer encapsulation.

Physical Procedures

Secondary forces between the particles and the modifier, like van der Waals, hydrogen, and electrostatic forces, are frequently produced by physical treatment (by covering the filler with a low molecular weight surfactant or a high molecular weight polymer). A lengthy aliphatic chain and one or more polar groups are both components of a surfactant. The fundamental idea behind surfactant treatment is the electrostatic interaction-driven preferential adsorption of the surfactant's polar group onto the fillers' high energy surfaces. Under specific conditions, ionic bonding can develop. Regarding CaCO_3 treated with stearic acid, a representative example can be given [20]. It has been demonstrated that the molecules of the surfactant and the filler surface formed ionic bonds. The shield consisting of C18 alkyl chains on the filler surface therefore had a lower surface energy. One further type of efficient surface modifiers for calcium carbonate fillers is alkyl dihydrogen phosphate, which contains functional groups like olefine, chloro, methacryloxy, and mercapto [21]. It has been demonstrated that alkyl dihydrogenphosphate molecules can interact with CaCO_3 to form a calcium phosphate salt that is dibasic. The products are deposited on CaCO_3 and provide it a degree of hydrophobicity, as seen by the mineral oil's excellent dispersibility [22]. Additionally, the surface area of the treated particles always increases as a result of treatment with alkyl dihydrogenphosphates. This can be attributed to calcium salt production or surface erosion.

Using preformed or in situ produced polymers to enclose inorganic small particles is another method of physical treatment. A macromolecular dispersant called a hyperdispersant has been employed successfully in the first instance. Polymeric dispersants are made up primarily of two elements, like surfactants. One is a functional group, such as $-\text{OH}$, $-\text{NH}_2$, $-\text{NR}_3^+$, $-\text{COOH}$, $-\text{COO}^-$, $-\text{SO}_3\text{H}$, $-\text{SO}_3^-$, and $-\text{PO}_4^{2-}$, which aids in anchoring the dispersants to the particles surface by

hydroxyl and electrostatic bonding. The second type of macromolecule is one that can be broken down into smaller pieces and dispersed in a variety of low- to high-polarity media, such as polyolefine, polyester, polyacrylate, and polyether. The following benefits over conventional surfactants when using hyperdispersants:

They cannot easily be desorbed because (i) they are more tightly tethered to the surface of the particles than surfactants are, and (ii) the long polymeric chains can more effectively prevent the reagglomeration of the particles. It should be emphasized, however, that due to their lengthy molecular structure, hyperdispersants can only enclose nanoparticle agglomerates and can scarcely diffuse into the agglomerates.

In comparison, encapsulation using polymers synthesized in-situ might result in more equal coverage. The polymer can be created by solution polymerization or emulsion polymerization in the presence of nanoparticles. Vinyl monomers were radical polymerized by Yoshinaga and colleagues [23] to create core-shell SiO_2 particles. Due to the negative charges that the surfaces of silica colloid particles have on the electric double layer, the initiator 2, 2'-azobis (2-amidinopropane)dihydrochloride can be concentrated on the silica surface and produce free radicals. To make the attachment of the macromolecular chains to the particles easier, a solvent that is less miscible with the polymer was utilized. Applying the emulsion polymerization technique allows for the encapsulation of small inorganic particles by a polymer layer [24]. Due to the monomer's adsorption on the surface of unaltered particles, polymerization largely takes place there, with further polymerization occurring in the layer where the monomer was adsorbed. It is essential for the creation of the polymer shell that the particles first create an initial hydrophobic layer on their surface, which can be accomplished by pretreating the particles with non-ionic surfactants, amphiphilic block copolymers, etc.

Chemical Techniques

Because the covalent attachment modifier prevents its desorption from the particles surface, its effect on surface modification based on chemical interaction is more substantial than that of physical approaches. Additionally, the addition of modifiers containing reactive groups

can be used to fine-tune the interfacial interaction in composites with chemically prepared nanoparticles. This aids in creating chemical bonding between the fillers and the matrix polymer. Coupling agent treatment is the most common and straightforward method among those that are accessible. In order to increase the adhesion between inorganic fillers and organic matrices, a range of coupling agents have been used, including silane, titanate, and zirconate. There is a lot of literature on coupling agent effects and response processes for microfillers, but not as much for nanoparticles. The characteristics of fumed silica covering with various silane types were explored by Gunko et al. [25] from both experimental and theoretical perspectives. The connected silanes can reduce silica's hydrophilic characteristics to various degrees, depending on the pretreatment circumstances. However, it should be noted that using straightforward physical and mechanical methods, it is difficult to achieve the uniform surface covering of the coupling agent on nanoparticles. There are certain benefits of grafting macromolecules onto inorganic particles as opposed to linking or low molecular surfactant modifications. By carefully choosing the species of graft monomer and the graft conditions, the polymer grafted particles can have the appropriate properties, allowing the interfacial properties between the treated nanoparticles and the matrix to be precisely controlled when creating nanocomposites. Additionally, the weak nanoparticle aggregates become more robust due to the formation of a nanocomposite microstructure made up of the nanoparticles, grafted and ungrafted (homopolymerized) polymer, and other components. In this area of expanding interest, numerous studies have been conducted with the goal of enhancing the nanoparticles' compatibility with polymers and their ability to disperse in solvents. The two main methods for graft polymerization are as follows:

(i) Grafting from is the process by which active substances (often known as initiators or comonomers) are converted into monomers and then polymerized. (ii) Grafting to refers to the reaction between ready-made polymers with reactive end groups and the functional groups on the particles.

The latter helps to regulate the grafting polymer's molecular weight. But generally speaking, the former method is superior to the latter since polymers have a harder time penetrating particle agglomerations than monomers do. Additionally, using the "grafting to" procedure would result in a polymer coating on the particles, which would prevent the attachment of other polymers.

There are two methods for starting polymerization from the particles when using the "grafting from" technique. One is surface modification using polymerisable groups, for as by treating surfaces with silane (Figure 1). Surface modification using initiating groups is the alternative (Figure 2). Comparatively, the former is simpler, whereas the latter favours grafting efficiency and a larger grafting percentage. Espiard and Guyot[26] first applied methacryloxy propyl trimethoxysilane (MPTMS), a silane coupling agent with a double bond, to the surface of silica nanoparticles. After that, polyethylacrylate was grafted onto the modified silica in an emulsion solution. In order to graft polystyrene (PS) and poly(styreneblock-isoprene) onto nanosilica (~ 12 nm), Oosterling et al.[27] first used t-butyllithium to initiate immobilized double bonds on the silica surface before adding the monomers to the reaction suspension. Similar techniques were used by Rong et al. [28,29] to graft PS and polyacrylamide (PAAm) onto nanosized alumina and silicon carbide in solution.

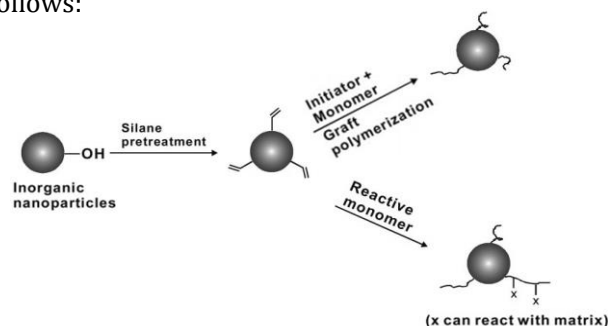


Figure 1: Diagram of a polymer that was grafted from nanoparticles using the double bonds on the particles

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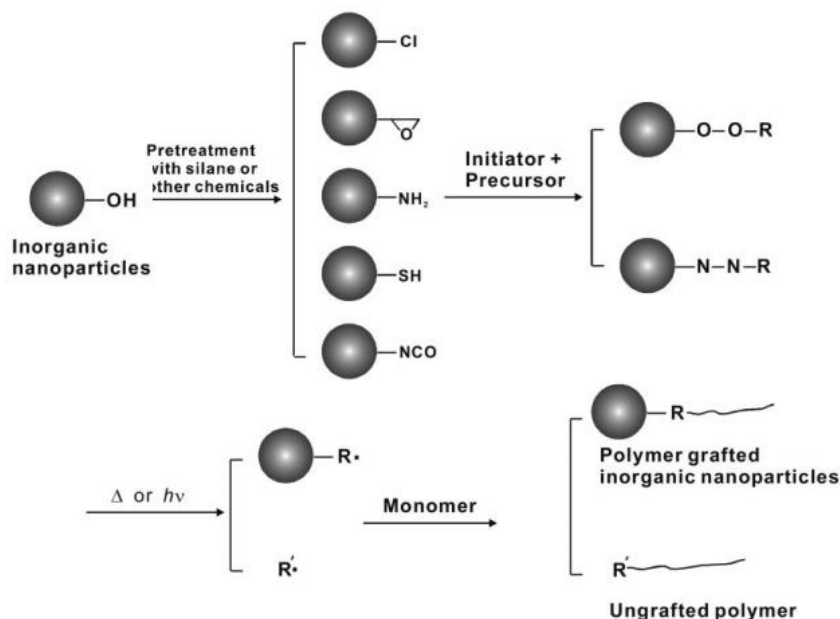


Figure 2: Using initiators on the particles, a polymer is grafted from nanoparticles in this schematic picture

Polyglycidylmethacrylate (PGMA) was also used to graft the same SiC nanoparticles into an emulsion without surfactants [30]. Because there are more SiC nanoparticles and they have a hydrophobic surface, it has been determined that the loci for emulsion polymerization are on the surface of these nanoparticles. The effectiveness of the PGMA graft and encapsulation around the nanoparticles depends in large part on this reaction process.

Different processes, including free radical, cationic, [31] ring opening metathesis polymerization (ROMP), [32] atom transfer radical polymerization (ATRP), [33, 34] and anionic polymerization, can lead to surface initiated polymerization [35, 36]. Tsubokawa and colleagues [37-42] examined the systematic grafting of different polymers onto ultrafine silica and carbon black by polymerization starting from the initiating groups previously placed into the surfaces. They have performed polymerizations of several monomers that are begun by potassium carboxylate, acylium or benzylium perchlorate, peroxyester or azo groups that have already been added to ultrafine silica, and radical, cationic, and anionic grafts, respectively. For instance, silica particles were treated with thionylchloride to produce surface Si-Cl groups, which were then used to react with tert-butylhydroperoxide or diisopropylbenzene hydroperoxide to create the initiating groups. This method affixed a surface-anchored peroxide initiator to the surface.

Typically, two methods are used to add the azo groups to the surface of untracined particles. One involves the interaction between 4, 4'-azobis(4-cyanopentanoic acid) (ACPA) and surface epoxide groups that are introduced when the particle is treated with c-glycidoxypropyltrimethoxysilane (GPS) [38]. The second involves ACPA's reactivity with surface isocyanate groups that are introduced when the particle is treated with toluene diisocyanate (TDI) [39]. Azo initiator can also be immobilized onto silica substrates by means of an amide bond generated between ACPA and an amine-containing silane coupling agent on the silica [40].

In addition to chemical grafting, γ irradiation can also be used to achieve graft polymerization onto nanoparticles. This method is preferable than chemical grafting in terms of graft uniformity, pollution control, and procedure simplification. The surfaces of the nanoparticles both outside and inside the agglomerates are equally stimulated by the high energy radiation. As a result, throughout the agglomerates, the low molecular weight monomers are permitted to interact with the nanoparticles' active sites [43]. In more recent research, it was discovered that, under relatively benign conditions, different polymers can be sequentially grafted onto the surface of silica nanoparticles (at air atmosphere) [44]. The nanoparticles' highly active surface may be the essential factor. Additionally, the reactivity of various silica

varies. Performance-wise, precipitated silica outperforms fumed silica [45]. Another intriguing finding is that even though some metal nanoparticles, like those of iron, nickel, and cobalt, contain few hydroxyl groups, they can still be grafted with polymers when exposed to radiation[46], suggesting that the reactive site may either be the hydroxyl group or the surface structural flaw. In fact, the silane's introduction of double bonds to the surface of the particles aids in the increased grafting percentage [47].

Influences of Interfacial Interaction on Polymer Nanocomposites' Performance

Composition, structure, interfacial interactions, and component characteristics all have a role in determining the properties of composite materials. The volume and qualities of interphase determine its impact. In other words, the amount of the material bonded in the interphase and the strength of the contact affect the final properties of the composite. Accomplishments in the development of nanocomposites would result from understanding the significance of the primary elements controlling interfacial adhesion and proper surface modification.

The fundamental notion behind nanocomposites is the creation of a very wide contact between the nanoscale constituent parts and the polymer matrix. However, it is frequently difficult to achieve the desired big surface due to the homogenous distribution of the nanoparticles. Furthermore, there are additional issues in tuning interfacial interaction or component coupling. By analyzing a few chosen references, the topics crucial to this field's future advancement and the problems that need to be handled are discussed in the lines that follow.

Effects of the Coupling Agent, the Surfactant, and the Dispersant

By melting and mixing, Chan et al.[48] introduced nano- CaCO_3 (44 nm) to polypropylene (PP). The PP matrix's elastic modulus and impact strength increased as a result of the nanoparticles. Yield stress and tensile strength of the composites were shown to have the opposite impact. These two amounts declined as the volume fraction of CaCO_3 rose and were maximum for the neat PP. However, a

recent study by another group demonstrated that the addition of 0–15 phr nano- CaCO_3 particles simultaneously improved the yield strength, elongation at break, impact strength, flexural modulus, and Vicat softening temperature of polyvinyl chloride (PVC) [49]. Because of its non-polar nature, it is assumed that PP will interact with nanoparticles extremely poorly, whereas PVC composites should exhibit substantially stronger interfacial interactions. It is clear that a significant factor in the effectiveness of composites is the interaction between the particles and the polymer matrix.

Surface modification of fillers with surfactant or coupling agent, which has been commonly used in creating microcomposites, is likely to operate in nanocomposites to strengthen interfacial contact. However, there aren't any universal fixes. Each set of nanocomposites requires its own unique approach, which must be created. PP was combined with CaCO_3 (50 nm) using a twin screw extruder by Zhang and Hua [50]. A titanate coupling agent that is water soluble was used to pre-treat the nanoparticles. The coupling agent-treated particles show better dispersion in PP when compared to nano- CaCO_3 that has been treated with fatty acid salts. With the addition of 5–6 phr of coupling agent prepared nano- CaCO_3 , the impact strength of PP increased by 70% but the fatty acid salt treated nano- CaCO_3 composites only saw a 50% rise.

Additionally, homemade phosphonate surfactants were employed to alter nano- CaCO_3 particles. Similarly, it was shown that nano- CaCO_3 /PP composites treated with phosphonate had improved Charpy impact strength. The impact of a phosphonate coupling agent was assessed in a study by Ren and colleagues[51]. When compared to untreated nano- CaCO_3 , treated nano- CaCO_3 can boost tensile and impact strength while untreated nano- CaCO_3 had essentially no reinforcing effect. From the discussion above, it is feasible to draw the conclusion that a number of coupling agents, such as titanate, phosphate, and aluminate[52], support the enhancement of nanoparticle dispersion and mechanical properties of polymer composites.

While silane is frequently employed in the silica treatment process, it is typically not suited for CaCO_3 surface modification. Such research

on the effects of commercial dispersant (Efka) and silane coupling agent (thiocyanatopropyl-triethoxysilane) pretreatment of nano-silica on tensile characteristics of PP were reported by Kruenate et al.[53]. When PP and untreated nano-silica were combined, the tensile strength and modulus increased while the elongation to break dropped. In contrast, silane treatment is more successful in enhancing the tensile strength and modulus of PP/SiO₂ composites (Table 1).

In actuality, stearic acid was used to pre-treat all commercial nano-CaCO₃ particles. Lazzeria et al. [54] investigated the effects of stearic acid

treatment on the mechanical characteristics of composites made of precipitated CaCO₃ (70 nm) and high density polyethylene (HDPE) using a twin screw mixer-single screw extruder. Young's modulus and yield stress of HDPE composites increased when 10 vol.% nano-CaCO₃ was added to it, but impact strength significantly decreased. While the impact strength gradually rose, the use of stearic acid resulted in a modest reduction in the composites' Young's modulus and yield stress when compared to the uncoated CaCO₃ composites.

Table 1: Mechanical properties of PP composites with treated and untreated nanosilica [53]

Material	Elongation to break, %	Tensile strength, MPa	Modulus, MPa
PP	21.12 ± 2.43	36.96 ± 0.40	1178.29 ± 42.18
PP/2%SiO ₂	13.19 ± 1.11	40.21 ± 2.16	1376.46 ± 63.79
PP/2%SiO ₂ /Efka	9.14 ± 2.38	38.19 ± 1.44	1414.67 ± 10.40
PP/10%SiO ₂ /Efka	10.12 ± 1.06	33.60 ± 0.48	1603.31 ± 58.43
PP/2%SiO ₂ /silane	11.80 ± 1.31	43.54 ± 1.85	1524.20 ± 45.57

With the exception of the wet treatment method, the aforementioned surface pretreatment of nanoparticles is often performed by merely combining the particles with a coupling agent or dispersant in a higher speed mixer (in water). However, because of the extremely low weight ratio of the coupling agent to the nanoparticle agglomerates, it is difficult to establish uniform surface covering of the particles. An ultrahigh speed mixer has been

created to address this issue [55]. The mixer operates at a high speed (6000 revs per minute), has specialized rotors made to handle nanoparticles, and adds coupling agent using an atomizer. It was shown that using an ultrahigh speed mixer to incorporate nano-CaCO₃ modified by stearic acid significantly improved the mechanical properties of PP, especially the ductile qualities (Figure 3).

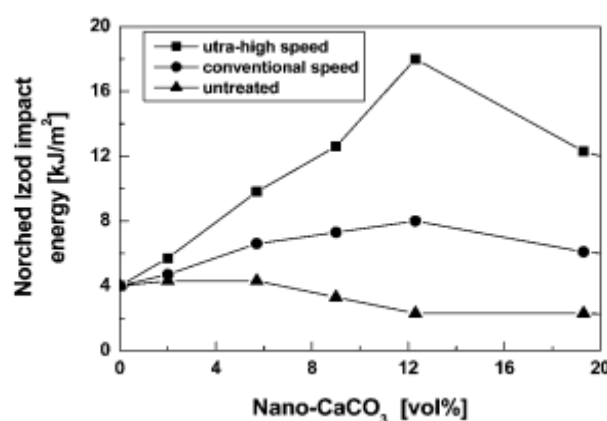


Figure 3: Impact energy of stearic acid treated nano-CaCO₃/PP composites as function of filler content [55]

The majority of the time, coupling agents or dispersants was used to prepare nanoparticles before mixing them with polymers. Since only the outside nanoparticles of the agglomerates can participate in the reaction, splitting the treated agglomerates during the following

compounding with polymers would expose the unreacted inside particles and cause them to reaggregate. The split nanoparticles must therefore be stabilized. Wu and Ye[56] started by adding HDPE to a two-roll mill that had been warmed to 150°C. Both CaCO₃ (40 nm) and

stearic acid were added to the fully melted polymer at the required composition. It was anticipated that stearic acid would act locally to activate the distributed nanoparticles in the polymer melt and so inhibit the production of bigger nanoparticle agglomerates. The viability of their concept was demonstrated by microstructural examination of the shattered surfaces of the composites. In other words, when the dispersion agent was used in the right dosage, hardly any nanoparticle agglomerates developed. Impact strength and elongation to break of HDPE were both improved by >100% for the composites made in this manner that contained 20 wt% of nano- CaCO_3 .

Impact of Compatibilizers and Macromolecule Encapsulation

Encapsulation with the high molecules, such as hyperdispersant, should enable greater interaction with polymer matrix for nanoparticles due to the higher interfacial

entanglement in comparison to coupling agent, surfactant, and dispersant that belong to small molecules. Li and Chen[57] discovered that the hyperdispersant CH-1A is more effective than the usual dispersant and stearic acid at enhancing the impact strength of PP/nano- CaCO_3 composites (Figure 4). For low density polyethylene (LDPE) composites enhanced with nano- CaCO_3 , similar results were described in Ref. [52]. The LDPE composite film filled with nano- CaCO_3 treated with macromolecular interfacial agent performed much better in terms of tensile strength and elongation at break than the composite containing nano- CaCO_3 treated with aluminate (Figure 5). High impact polystyrene (HIPS)/ TiO_2 (20 nm) nanocomposites were also reported by Zhang et al.,[58] exhibit comparable behavior. In addition to improving the composites' Young's modulus, tensile strength, and impact resistance, nano- TiO_2 content of < 5 wt% also increases their hardness.

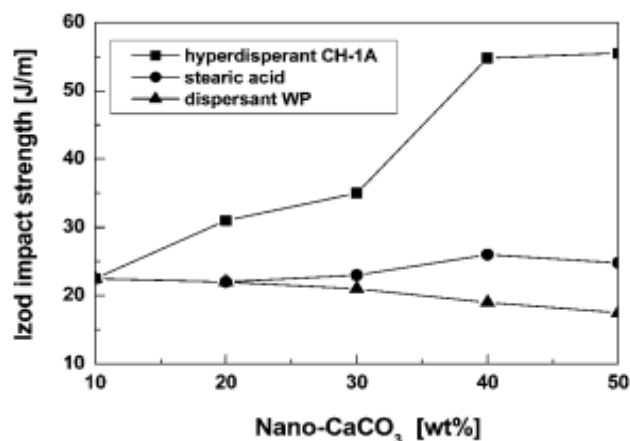


Figure 4: Effect of filler content on the Izod impact strength of nano- CaCO_3 /PP composites [59]

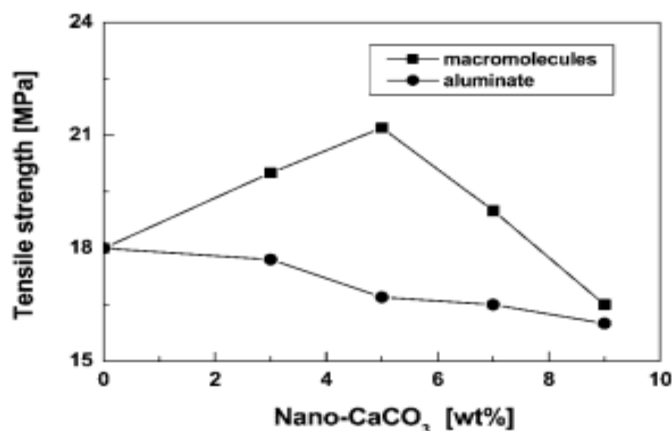


Figure 5: Tensile strength of composites made of nano- CaCO_3 and LDPE as a function of filler content [60]

By melt mixing, Wang and Yan [61] created PP/CaCO₃ nanocomposites in which the CaCO₃ nanoparticles had either undergone stearic acid pretreatment or been enclosed in poly(methyl methacrylate) (PMMA). By using in situ solution polymerization, nano-CaCO₃ was physically enclosed in PMMA. According to their findings, the PMMA-coated nano-CaCO₃ particles were rather evenly distributed throughout the PP matrix. However, the mechanical performance of PP composites containing either stearic acid-treated nano-CaCO₃ or nano-CaCO₃ enclosed in PMMA was comparable. When compared to clean PP, the impact strength of PP composites was improved by a factor of 2. In a different study on PVC/nano-Sb₂O₃ nanocomposites, the researchers in situ polymerized methylmethacrylate (10 weight percent based on nano-Sb₂O₃ particles) to modify the surface of the nano-Sb₂O₃ [62]. The resulting PMMA shell wrapped on nano-Sb₂O₃ particles increases the contact between the nanoparticles and the PVC matrix, causing the dissolution of the nanoparticle aggregates and improving the dispersion of the nanoparticles in PVC. Thus, at a loading of 2.5 wt%, the treated nano-Sb₂O₃ particles stiffen, reinforce, and toughen PVC matrix to their greatest extent. Wu et al. [63] produced a CPE/nano-CaCO₃ (40 nm, titanate pretreatment) masterbatch and used chlorinated polyethylene (CPE) as an interfacial modifier for PVC/CaCO₃ nanocomposites. A CPE layer was found to be encapsulating the nano-CaCO₃ particles in the PVC matrix, according to morphology analysis. Due to the addition of CPE, the notched Izod impact strength of the nanocomposites significantly improved and reached a high value of 745 J/m.

In addition to surface-modifying nanoparticles, compatibilizers like PP grafted with maleic anhydride (PP-g-MAH) have been shown to enhance the mechanical properties of nanocomposites. In PP/nano-CaCO₃ composites, Zhang et al. [64] investigated the impact of PP-g-MAH as an interfacial modifier. Through the process of solid phase graft copolymerization, PP-g-MAH was created. The PP/nano-CaCO₃ composites' notched impact strength (Figure 6), flexural strength, and tensile strength were all increased by the addition of PP-g-MAH. Reactive compatibilization of PP-g-MAH and CaCO₃ was demonstrated using Fourier transform infrared (FTIR). A more difficult manufacturing process to create PP/nanoCaCO₃ composites was disclosed in Ref. [65]. Nano-CaCO₃ that had been

treated with the silane coupling agent KH-570 was encased by polymerized organic monomers in an emulsion, and PP-g-MAH was made using a related technique. Melt mixing was used to create PP, PP-g-MAH, and encapsulated CaCO₃ based nanocomposites. It was discovered that the composites' Izod impact strength increased over plain PP by a factor of two. The optimal composition was found to be filler/encapsulating polymer 10: 1, PP-g-MAH 10 wt%, and CaCO₃ 5 wt% by cross-experiments for the impacts of the filler/encapsulating polymer ratio and the contents of PP-g-MAH and filler. When making a PP/nano-CaCO₃ masterbatch, Huang et al. [66] performed in situ graft polymerization of acrylic acid, which was subsequently diluted with PP by an extruder and injected into nano-CaCO₃/PP composite specimens. It is demonstrated that while impact strength varies in a reverse trend, this measure can significantly improve the tensile strength and modulus of the nanocomposites. In order to create melt mixed polyethersulphone (PES) composites, epoxy has been employed as a processing aid and a dispersing agent for fumed nanosilica [67]. Epoxy component was polymerized to regain its mechanical characteristics after dispersion in matrix polymer. It was discovered that the silanol groups of nanosilica and the polar groups of epoxy had a powerful interaction that aided in dispersing the particles into PES. Improvements over neat PES were seen in barrier resistance to solvent diffusion, including that of methylene chloride, heat deflection temperature, and impact strength.

Effects of Reactive Compatibilization and Graft Polymerization

PP/silica grafted hyperbranched polyester nanocomposites were created by Xu et al. [68], with very intriguing findings. Hyperbranched polyester (Boltorn H20) was used to graft the silane 3-glycidioxy propyltrimethoxysilane (GPTS) prepared silica (30 nm), which was then ended by octadecylisocyanate (C19) groups after the graft process. The notched impact strength (Figure 7) and tensile strength of the composites were significantly improved by the addition of silica grafted modified Boltorn H20 (3-5 wt-%SiO₂). When silica grafted modified Boltorn H20 particles were added to PP composites, as opposed to PP/SiO₂ or PP/SiO₂-GPTS composites, the mould flow index of the PP composites increased. Agglomeration occurs naturally in nanocomposites because

nanoparticles are so tiny. It has been established that nanoparticle agglomerates typically lack the necessary strength[69]. As a result, the agglomerates would easily fracture under the applied force, resulting in the premature collapse of the entire composite. In light of this, Rong et al.,[70] carried out a special series of experiments involving SiO₂ nanoparticles in PP matrix with and without the addition of grafting polymers (either PS or PMMA), which have the capacity to promote nanoparticle dispersion, improve the interfacial interactions between the particles and matrix,

and in particular increase the agglomerates strength. The grafting polymer that surrounds SiO₂ aids in boosting the nanocomposites' tensile strength, but almost has no impact on their elastic modulus. In comparison to clean PP, the strain to failure was lower in the untreated systems. However, in the treated system, the parameter went up by up to 2 vol % with the nanoparticle content. The impact strength of PP composites with grafted nano-SiO₂ is 3–4 times greater than that of pure PP at the same filler loading.

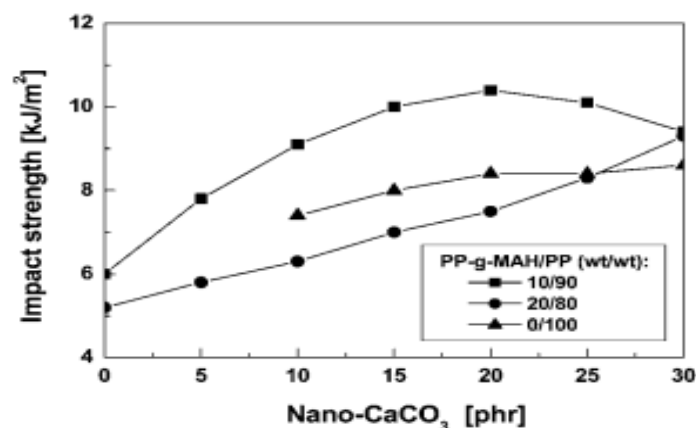


Figure 6: Impact strength of nano-CaCO₃/PP composites with a notched surface as a result of the PP-g-MAH [64]

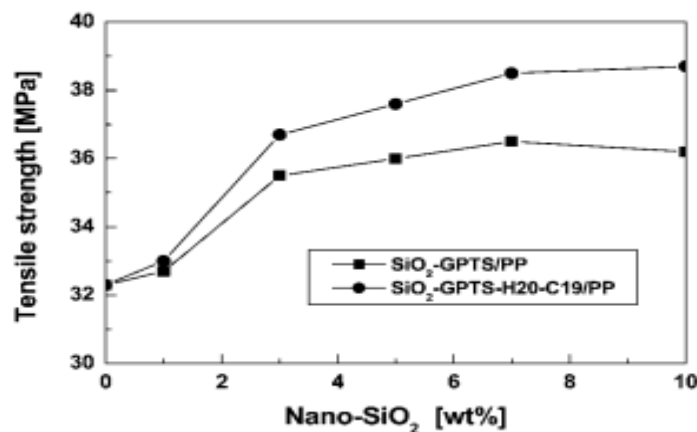


Figure 7: Tensile strength as a function of filler content for nano-SiO₂-GPTS/PP and nano-SiO₂-GPTS-H2O-C19/PP composites[68]

The idea of grafting polymer onto nanoparticle agglomerates to reinforce them also applies to the creation of antiwearing nanocomposites. Shi et al., [71] comprehensive study examined the sliding wear characteristics of epoxy incorporating nano-Al₂O₃ particles with various pretreatments. In this study, acrylamide, styrene, and ethylacrylate were each grafted onto nano-Al₂O₃ using the process of radiation polymerization. It has been

established that the polyacrylamide amide groups participate in the curing of epoxy resin, significantly improving the interface interaction [72]. It is thought that encasing nano-Al₂O₃ in polystyrene or polyethylacrylate can greatly improve the particles' hydrophobicity and, consequently, their compatibility with the polymer matrix. According to Figure 8, adding nano-Al₂O₃ particles significantly lowers the high wear rate of unfilled epoxy caused by the three-

dimensional crosslinking network. This is especially true when treated nano- Al_2O_3 is used [73]. Additionally, the nanoparticles lower the epoxy's frictional coefficient, and composites made with nanoparticles that have been treated have lower frictional coefficients. For instance, the frictional coefficient of Al_2O_3 -c-PAAM/epoxy is 0.35, which is equivalent to 61% of the value of epoxy that isn't filled and 69% of the value of nano- Al_2O_3 /epoxy that hasn't been treated.

Recently Friedrich et al., [74] combined reactive compatibilization with a chemical graft method to nanoparticles to generate nanocomposites. Specifically, reactive groups (epoxide or amine groups) were first added to nanoparticles by chemical graft polymerization, and then melt mixed with PP at various ratios. In the meantime, variously functionalized polypropylene (PP), such as maleic anhydride-grafted polypropylene (PPMAH), amine-functionalized polypropylene (PP-NH₂), and epoxide-functionalized polypropylene (PP-GMA), were added in order to strengthen the interfacial adhesion and improve the mechanical properties of the filler/matrix by forming chemical bonds as it relates to the manufacture of composites. The ability of

reactive compatibilization to improve mechanical characteristics, interfacial adhesion, and nanoparticle dispersion have been demonstrated. For greater wear resistance, such reactive compatibilization has also been effectively used in SiC/epoxy nanocomposites [75]. In this study, soapless emulsion polymerization was used to chemically graft PGMA onto pretreatment nano-SiC particles made of silane. The grafted versions of the nano-SiC particles showed better dispersibility when compared to untreated nano-SiC particles. Additionally, through a reaction between the epoxide groups and the curing agent, they are able to create chemical bonds between the fillers and the epoxy matrix. Due to these benefits, epoxy composites filled with PGMA grafted nano-SiC particles have higher sliding wear resistance and lower frictional coefficient at low filler loading than a SiC nanoparticle/epoxy system that has not been treated. In contrast to Al_2O_3 -g, Al_2O_3 -c-PAAM refers to the grafting of Al_2O_3 nanoparticles by PAAM without the removal of homopolymers produced during graft polymerisation. Al_2O_3 nanoparticles are only found with grafting polymers that are not extractable, according to PAAM.

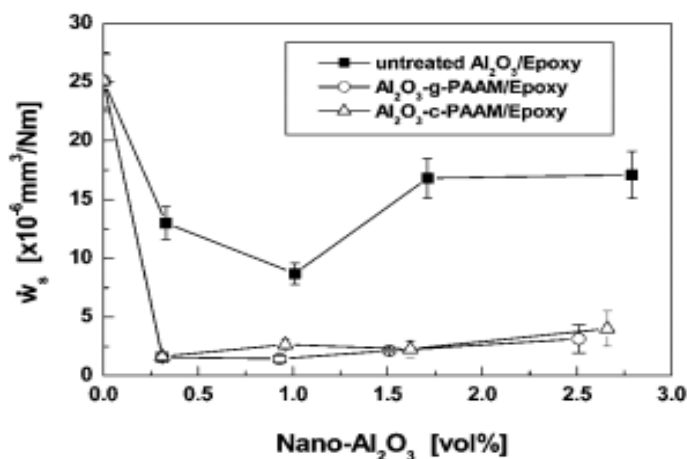


Figure 8: Specific wear rates of clean epoxy and its composites containing nano- Al_2O_3 before and after surface modification (testing configuration: block on ring) [73].

Models Using Micromechanical

Only a few micromechanical models have been created that are in agreement with the behavior of nanocomposites, in contrast to the experimental investigations in the field. The elastic response of polymer nanocomposites was predicted by Chaberta et al., [76] using a mean field technique or a discrete simulation. A self-consistent scheme that introduces a third rigidifying phase of matrix immobilized on the

surface of filler particles and a discrete model of sphere assembly that takes into account the local contacts between filler/filler and matrix/filler were the two micromechanical models that were proposed to account for the short range interactions between filler particles. The two models' physical foundations are identical, but they were approached in different ways: in the homogenization scheme, the thickness of the interphase (or effective

volume of the filler) is the key parameter, whereas in the discrete simulation, the surface of contact (or interaction forces between filler particles) governs the mechanical behavior. As a result, in the self-consistent model, the effective proportion of filler must be significantly increased to account for the observed reinforcement, in contrast to discrete simulation, where a small volume of immobilized matrix enclosed between filler particles significantly increases the modulus. The aforementioned models were validated using a nanocomposite made of submicronic stiff PS particles scattered randomly in a pliable polybutylacrylate (PBA) matrix. The discrete model underlined the significance of filler-filler interactions on the reinforcement over the percolation threshold in addition to the good agreement with the experimental results.

The interactions between polymers and nanoparticles as well as the structure and dynamics of the polymer melt containing nanoparticles were covered by molecular dynamic simulations performed by Starr et al.[77]. They discovered that the chains are lengthened and flattened close to the nanoparticle surface; as a result, the glass transition temperature (T_g) of the melt can be changed to either higher or lower temperatures as a result of the gradual alteration in the dynamics of the polymer as it gets closer to the nanoparticle surface. That example, it is expected that attractive contacts will increase T_g whereas very weak interactions can reduce T_g as a result of changes in surface dynamics. The bonded polymer area will likely exhibit attractive monomer-particle attractions if it is narrow (has very few layers). T_g will likely fall for bonded polymers with a greater thickness (several layers), where the outside layers of the monomer relax much more quickly than the inner monomers.

Polymer Nanocomposites Applications

Solar cells: The amount of non-renewable resources, such as fossil fuels, that are used up each year increases with the level of energy consumption in the modern world. Hence, the importance of renewable energy is crucial. The field of nanotechnology is one that is rapidly developing and has a wide range of uses. As a result of their unique features, polymer nanocomposites are a new breed of difficult materials with a variety of uses. Polymers that are encased in nanometals can boost the

efficiency of solar cells. More precisely, it is able to absorb solar energy. The optimum alternative for solar cells has a quantum yield value of more than one, which are plasmonic metal nanoparticles. Composites made of metal and polymers function like a metal-dielectric combination. Surface Plasma Resonance is the name for the collective oscillations of electron clouds (SPR). This SPR was produced at the interface between the dielectric matrix and the metal nanoparticles. This oscillation is caused by the interactions between metal nanoparticles and incident light [9].

Sensors: Sensors are employed to identify environmental changes. They therefore have uses in both biological and environmental contexts. Due to its SPR excitation, polymer nanocomposites serve as sensors [78]. The SPR is one of the most crucial features of metal nanoparticles. Interdisciplinary applications result from the blending of metal and polymer characteristics. The performance of sensors can be enhanced via polymer-metal composites. The SPR oscillations give the sensor its sensing ability. As a result, these materials are employed in biosensors as virus and cancer molecule detectors [79]. In sensor-oriented applications, polymer nanocomposites are utilized to analyze biological and environmental problems.

Thin Films: Today, a variety of fabrication methods are employed to create thin films. Metal nanoparticle thin film fabrication from a single material is challenging. The production of encapsulating thin films and a substrate for metal-nanocomposites is assisted by polymers. When creating or modeling nanodevices polymers with metal nanoparticles are crucial. In comparison to metal nanoparticles alone, a combination of metal nanoparticles and polymers can yield films of superior quality. These films are divided at the micro-phase level between the polymer matrix and the metal nanoparticles. Strong covalent bonds and light-weight molecular ligands are used to link polymers and nanoparticles together. Thin films are employed in the SPR-based fabrication of the sensors [79].

Microbial Applications: Silver nanoparticle and poly vinyl alcohol thin film composite exhibits antibacterial action. They can be reused, and monitoring the film is simple [80]. Polymers like PVA exhibit antimicrobial properties. By combining inorganic nanoparticles with metals like copper and silver

as well as with antibacterial qualities, we can improve them. The release of metal ions into the microbial cell aids in the investigation of the antimicrobial effects. Metal ions that have been discharged have the potential to harm a cell's microbiological environment [81-84]. An excellent photobactericidal activity is provided by the TiO₂ polymer matrix with inorganic nano-fillers.

Conductance: Composites made of polymer and metal nanoparticles can enhance their conductivity in order to incorporate it into electronic circuitry. The electro-spinning technology can be used to create nanofibers. When polymers with conductivity are used to create nanocomposites, conductivity can be increased. These polymers are utilized because to their light weight, low cost, flexibility, and excellent solvent reactivity. They fall within the category of bio-medical electro-active biomaterials [82-84]. The presence of free electrons determines the material's conductivity, a polymer that conducts electricity due to free electrons in its atoms. Conjugated double bonds are a type of polymer structure where single and double bonds alternate. Due to oxidation or reduction, a polymer with conjugated double bonds might have conductive properties. Conductivity can be improved by doping with various particles [85]. Semiconductor polymers are utilized as photodiodes in electroluminescence applications, Light Emitting Diodes (LEDs), and in the field of energy-saving bulbs [85].

CONCLUSIONS

A review of a sample of recent, representative literature was conducted to highlight the key issues surrounding surface modification of nanoparticles and their impact on the mechanical behavior of nanocomposites in an effort to better understand how nanosized inclusions affect the properties of polymer nanocomposites. It was found that coupling agents, such as surfactants and dispersants, typically work well to improve the mechanical performance of polymer nanocomposites but are rarely able to produce appreciable improvements in overall characteristics. This makes sense given how challenging it is to apply coupling agent evenly all across nanoparticles. In comparison, macromolecules that encase nanoparticles perform better than tiny molecules because of the stronger interface entanglements. Chemically inducing graft

polymerization from the surface of nanoparticles proves to be an effective method for strengthening the weak nanoparticle agglomerates. Reactive compatibilization between the grafted nanoparticle and the matrix polymer would further aid interfacial adhesion, particularly when reactive groups are added to the grafting chains. Agglomeration of nanoparticles can generally be prevented when creating nanocomposites with conventional compounding facilities and processes, despite the use of a variety of treatments and modification approaches. For the time being, every effort must be taken to reduce it as much as possible. It is important to note that the mechanical properties of polymer nanocomposites depend on a variety of factors, including crystallinity, the amorphousness of the polymer matrix, and multimorphologies, in addition to interfacial properties. The positive effect of nanoparticles in polymer matrix may be fully exploited on the basis of systematic consideration of all possibilities (in actuality, many techniques for modifying nanoparticles, as intensively reviewed previously, have not yet been applied to the manufacturing of nanocomposites). This would result in much higher properties than those ever reported.

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